

Alexis CHALLANDE

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PROFESSIONAL SUMMARY

Security Researcher (Ph.D.) specializing in scalable static analysis and software supply chain integrity. Architect of critical security components for Sigstore. Expert in leveraging Large Language Models (LLMs) and agentic workflows to automate vulnerability discovery across massive, complex codebases. Committed to hardening global software ecosystems through high-impact research and engineering.

PROFESSIONAL EXPERIENCE

Trail of Bits, Remote • Security Engineer

09/2023 - Present

- o Designed and deployed scalable infrastructure to evaluate the performance of LLM-based offensive agents by deploying scenarios in Azure.
- o Performed critical vulnerability analysis on an upstream open-source project and a client's fork, identifying security flaws through detailed patch analysis, exploit generation, and variant analysis across both codebases. Automated the entire research workflow using GitHub Security Lab Taskflow to reduce reaction time.
- o Discovered and published findings on multiple Denial of Service recursion vulnerabilities in large-scale projects at [DistrictCon 0](#) using a custom CodeQL query..
- o Found several vulnerabilities in open source projects by leveraging various AI agents workflows (custom harnesses, variant analysis).
- o Designed and built the RFC3161 timestamping client that powers package attestations across the Python packaging ecosystem (PyPI, sigstore-python, pyca/cryptography), and contributed to PEP 740 adoption in PyPI.

Quarkslab, Paris • Security Engineer (Ph.D. candidate)

10/2018 - 12/2022

- o Conducted doctoral research on the detection of 1-day vulnerabilities within Android phones.
- o Performed security audits by reverse-engineering Android applications.
- o Developed an IDA Pro plugin (Quokka) for automated exports generation of arbitrary binaries.
- o Designed and implemented specialized security tools for the Android Open Source Project (AOSP), enhancing vulnerability detection capabilities within a large open-source codebase.

ANSSI (French Cybersecurity Agency), Paris • Intern

03/2018 - 08/2018

- o Developed and implemented an IDA Pro plugin using symbolic execution to automatically detect and analyze cryptography usage in binary code.

EDUCATION

Ph.D. in Computer Science • Ecole Polytechnique, Palaiseau

2022

M.Sc. in Computer Science, Digital Security • Eurecom, Biot

2018

SKILLS

AI: LLMs (GPT, Claude), Agentic Workflows (GitHub Taskflow)
Programming Languages: Python (expert), C (intermediate), C++ (intermediate), Rust (intermediate)
Reverse Engineering: IDA, Ghidra
Development Tools: git, CMake, Docker, GitHub, GitHub Actions
Cloud: Azure/AWS, Terraform, CDKTF

PUBLICATIONS

Denial of Service using recursions: not even Rust is going to save you • SSTIC, Rennes 2026
Shipping post-quantum cryptography to Python • Trail of Bits blog 2026
We hardened zizmor's GitHub Actions static analyzer • Trail of Bits blog 2026
Making PyPI's test suite 81% faster • Trail of Bits blog 2025
Low-Effort Denial of Service with Recursion • DistrictCon, Washington DC 2025
Don't recurse on untrusted input • Trail of Bits blog 2025
Quokka: A Fast and Accurate Binary Exporter • GreHack, Grenoble 2022
Towards Patch Detection using Binary Only Semantic Signatures • HAL 2022
Building a commit-level dataset of real-world vulnerabilities • CODASPY, Baltimore 2022
Commit Level Vulnerability Dataset • Quarkslab blog 2022
Exploitation du graphe de dépendance d'AOSP à des fins de sécurité • SSTIC, Rennes 2021
An Experimental Study of Different Binary Exporters • Quarkslab blog 2019

DISCLOSURES

GHSA-q729-696q-g9pq / GHSA-q8qp-67f9-wr3f • SurrealDB 2026
CVE-2026-23896 • Immich 2026
CVE-2025-33238 / CVE-2025-33254 • NVIDIA 2025
CVE-2025-4565 • Google 2025
CVE-2024-52981 • Elastic 2024
CVE-2024-52980 • Elastic 2024
CVE-2024-58103 • Wire 2024
RUSTSEC-2024-0437 • rust-protobuf 2024
CVE-2024-47072 • XStream 2024
CVE-2024-7254 • Google 2024